Aurora Forecast Platform

Real-time, Hyper-local Space Weather Intelligence

Team: **DOMINATORS** | Orion Astrathon 2026

GitHub: **GitHub Repository**
Website: **Live Deployment**

Team Details

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- **Github Repository:** https://github.com/InfinityxR9/Orion_Astrathon_26
- **Live Website:** https://thedominatorsbackend.onrender.com/

1. The Problem Statement

> **The "Kp" Flaw:** The standard 3-hour Kp index systematically misses 10-30 minute intense auroral substorms.

> **Fragmented Tools:** Photographers must manually cross-reference space weather, local cloud cover, and light pollution.

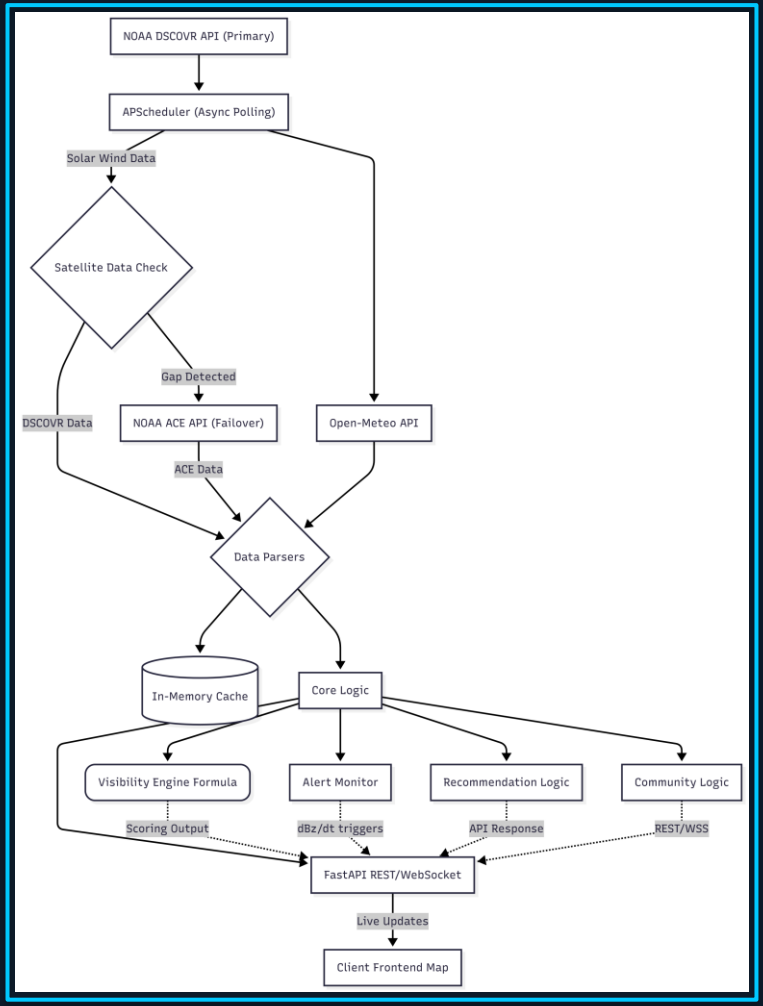
> **The Objective:** Build a real-time platform that answers: "Can I see aurora from this exact location right now?"

2. Methodology

- **Product Methodology:** User-first decision flow (location -> visibility score -> field recommendation).
- **Scientific Methodology:** Treat aurora visibility as a multiplicative observability filter, not an additive bonus.
- **Engineering Methodology:** Async ingestion separated from request handling with multi-layer fail-soft caching.

3. Final Architecture Diagram

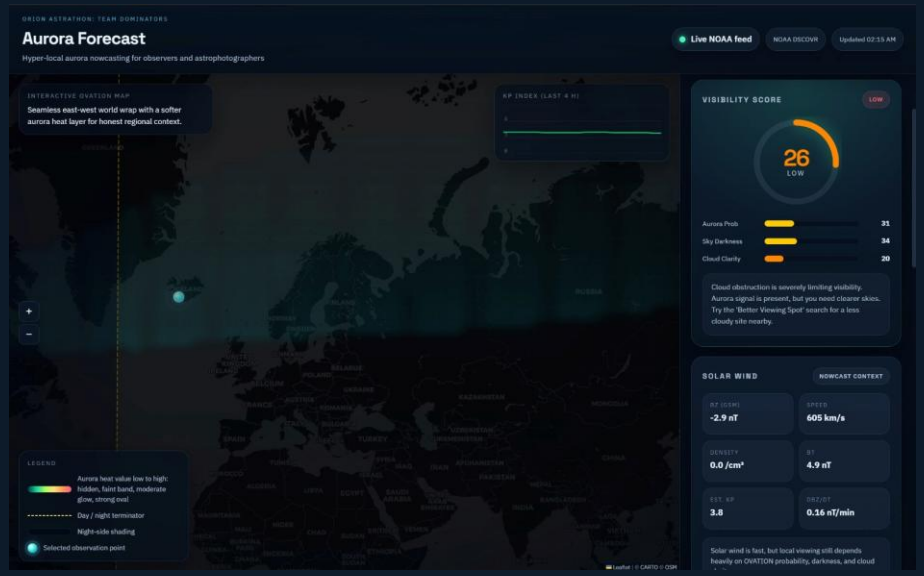
> Data flows: NOAA DSCOVR (Primary) → APScheduler → Satellite Data Check → DSCOVR/ACE Failover → Data Parsers → In-Memory Cache & Core Logic → FastAPI REST/WebSocket → Client Frontend Map.



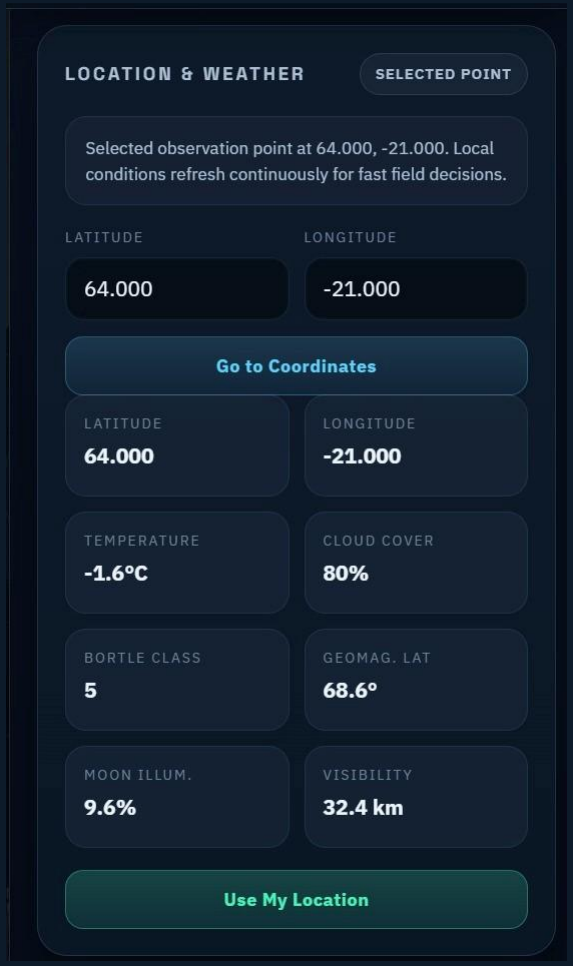
Core Features (The Dashboard)

> **Live Map:** Dynamic OVATION heat layer and day/night terminator.

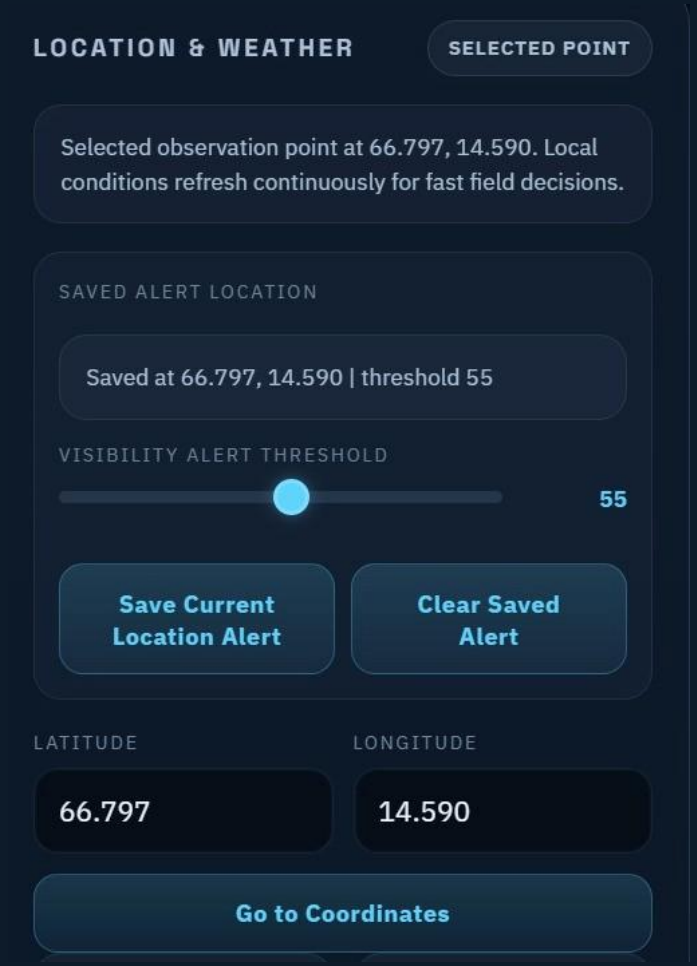
> **Alert System:** Configurable push-notifications when visibility scores exceed user-set thresholds at saved locations.



Main Page showing Probability heat map (Faded due to low Aurora probability), The overall metrics and visibility score, and Information about solar winds



Part of Right panel showing Coordinate and Weather Information



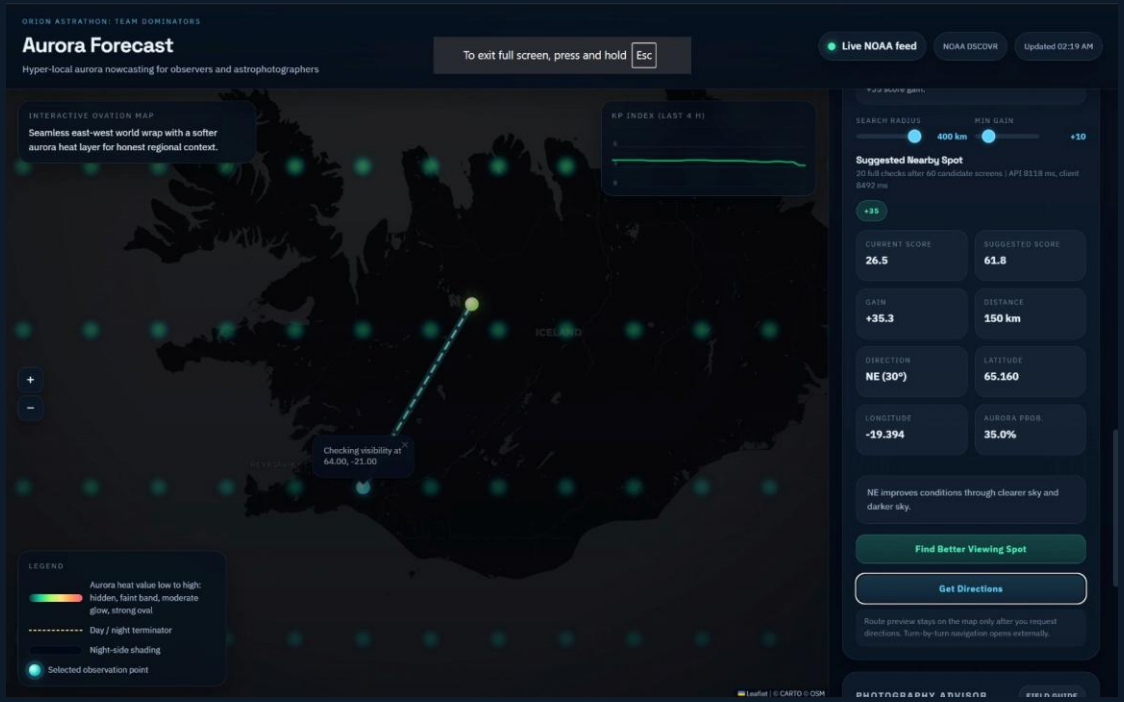
Alert system

Stretch Goals (Intelligence)

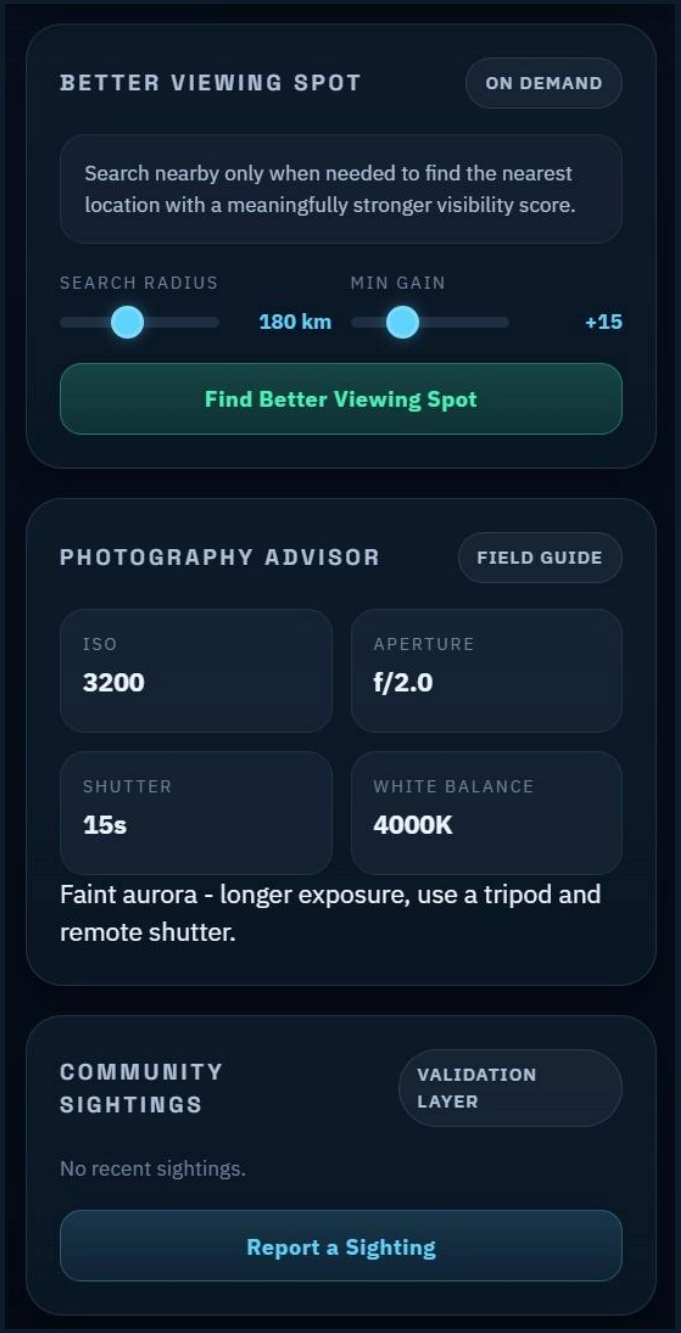
> **Nearby Recommendation:** Using the better spot feature to get directions to the near better view spot as per user selected parameters.

> **Photography Advisor:** Recommends ISO, aperture, and shutter speed based on the live score.

> **Community Sightings:** Algorithm to find the better viewing spot with user-selectable search radius, minimum visibility score gain, and a Community Sightings Review section.



Using the better spot feature to get directions to the near better view spot as per user selected parameters



Algorithm to Find the better viewing spot with the user selectable search radius and Minimum visibility score gain option, and Photography Advisory as per current weather conditions, and a Community Sightings Review section

Offline Survivability (PWA)

> **PWA Integration:** Progressive Web App (PWA) Integration enables field use without 4G, or when the cellular data is very weak.

> **App Shell Caching:** Pre-caches the App Shell (HTML, JS, CSS).

> **Map Tile Cache:** Caches Map Tiles for immediate offline rendering.

> **Network-first Strategy:** Serves cached API snapshots (Network-first, Cache-fallback).

PWA makes use of the existing data rendered by the website already in the form of cache and using it efficiently in cases of internet time-outs.

4. Test Results & Integrity

~230 ms	PASS	PASS
Better Viewing Spot Latency	Primary API Failover Test	Malformed JSON Parser Tests

> **Latency:** "Better Viewing Spot" algorithm processes typical checks in ~230 ms via parallel fetching.

> **Resilience:** Successfully passed failover tests for simulated primary API outages.

> **Data Integrity:** Passed parser hardening tests against malformed NOAA JSON rows.

Visibility Score Engine (Metrics Detail)

$$\text{Visibility} = 100 \times A_norm \times \text{sqrt}(D) \times \text{sqrt}(C)$$

$$A_norm = \min(\text{Aurora_Probability} / 30,\ 1.0)$$

> **Aurora Normalisation:** A_norm = min(Aurora_Probability / 30, 1.0) — caps raw OVATION probability at 30% to stabilize local scoring.

> **Geometric Veto:** Using a geometric mean for Darkness (D) and Clouds (C) ensures thick clouds mathematically drop the score to zero.

> **Decision Quality:** Visibility score range adherence [0, 100] and component consistency.

> **Reliability:** Feed freshness lag tracking and data-gap detection rate.

Conclusion & Links

- **From Data to Decisions:** Transformed raw scientific telemetry into an instant, actionable field tool.
- **Ready for Deployment:** Codebase is complete, Docker-ready, and resilient to upcoming NOAA schema changes.

Thank You

Live Demo: Live Demo

GitHub Repo: GitHub Repository